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CHAPTER 1
INTRODUCTION & STARTING THE EXPEDITION

Welcome to the surface control station, Operator.

Before you, the terminal screen glows a dim green in the middle of the cold control room. Thousands of meters below, stretches the abyss—a place where sunlight never reaches, water pressure can crush thick steel in seconds, and a pitch-black silence hides unexplored ancient mysteries.

An unmanned submersible vehicle (ROV) is your eyes, ears, and hands in the perpetual darkness. Your task is simple but fraught with risk: dive into the ocean trench, collect valuable anomalous samples, and ensure the probe returns safely before the trench's inhabitants discover your presence.

1. Purpose of the Ocean Trench Expedition

As a remote operator, your mission is divided into three main focuses:

1. **Depth Exploration:** Guide the vehicle through the coral cliff passages and narrow troughs without hitting the rock walls.
2. **Valuable Sampling:** Search for biological anomalies (Biomatter), rare minerals (Rare Ore), and even the remains of previous expedition recordings (Black Boxes). Each sample will be converted into Research Points (RP).
3. **Survival Management:** Maintaining the physical integrity of the hull (Hull Integrity), saving battery (Power), and avoiding seabed predators that are attracted to the vibrations and sound of your engine.

2. Main Switch: Using the Data Link Button

Your control center relies on one main communication link located on the right panel of the screen, namely the DATA LINK button.

A. Launching the Ride (`DATA LINK [CONNECT]`)

- When a new game is opened, the ride is in Standby status on the surface dock.

- The button will light up with the text DATA LINK [CONNECT] in light blue.
- **How to Get Started:** Click the button once. The system will immediately establish a telemetry link, the mission clock will start, and the ROV will be lowered into the water column. The button text will change to red, reading DATA LINK [DISCONNECT], indicating that you are now in full control underwater.

B. Data Evacuation & Security (`DATA LINK [DISCONNECT]`)

- In the depths of the ocean, greed is the greatest enemy. Disconnecting early is the smartest survival tactic (Hit and Run).
- **When to Disconnect?**
  - When you have successfully secured 1–2 high-value samples and want to keep the points permanently.
  - When the hull of the vehicle experiences severe cracks due to impact or attack.
  - When the battery power is at critical threshold.
- **Emergency Extraction Method:** Click the red DATA LINK [DISCONNECT] button. The system will immediately disconnect the signal, locking the mission status to successful/survived, and all collected research points will be safely stored in the base's safe.

Important Operator Tips:

Hangar Door Locked While Diving: *Access to the ride upgrade menu (*ACCESS HANGAR) will be automatically disabled once Data Link is active. You can only modify or repair ROV parts while the vehicle is on the surface in a disconnected state (DATA LINK [CONNECT]). Make sure you're fully prepared before pressing the connect button!

CHAPTER½
ROV¤BASIC£CONTROL° ¤NAVIGATION

The tether vibrates gently as a cold current sweeps across your vehicle's titanium hull. At thousands of meters deep, there's no sky, no conventional compass, and no instant braking. The thick seawater exerts a significant drag—every thrust of the engine launches the vehicle with inertia, and every turn of the rudder requires anticipating a certain amount of maneuver to avoid the ROV's nose hitting the hard trench wall.

Piloting an ROV is not about speed, but rather the art of maintaining momentum and keeping your fingers steady on the console.

1. Adjust the Thruster Output

The main thruster is the power source for your vehicle's forward and backward motion. You can control it using a sliding lever on the control panel or using keyboard shortcuts.

A. Reading the THRUSTER OUTPUT Lever

- **Neutral Position (50%):**The engine is in an idle or stationary position. The vehicle receives no horizontal or vertical thrust.
- **Forward (51% – 100%):**The rotating propellers propel the vehicle forward while gradually increasing the diving speed.
- **Retreat (0% – 49%):**The propellers reverse direction, slowing the ride or pulling it back out of a narrow gap.

B. Using Keyboard Buttons (Fastest Method)

Moving levers with the mouse in a pinch can often be time-consuming. Use the following keyboard shortcuts for instant response:

- **[W] key:**Increases forward thrust by +5% per beat.
- **[S] key:**Reduces thrust or reverses the ride by -5% per tap.

Physics & Latency Notes: *Thrust commands are sent via fiber-optic cable with minimal transmission delay (signal latency). The engines take a fraction of a second to spin the propellers and build acceleration. Don't press buttons blindly until the spacecraft is already taking off!*

2. Controlling the Steering Direction

The rudder determines which direction the vehicle is headed before you engage the main thrusters.

A. Reading the STEERING Lever

- **CENTER (50%):** Steer straight ahead.
- **PORT (0% – 49%):** The vehicle turns its bow to the left side (Portside).
- **STBD / Starboard (51% – 100%):** The vehicle turns its bow to the right side (Starboard).

B. Using the Keyboard Steering Wheel Buttons

- **Press & Hold [A]:** Dynamically rotates the bow to the left (-10% per beat).
- **Press & Hold [D]:** Dynamically turns the bow to the right (+10% per beat).
- **Releasing the [A] or [D] Button:** Once the rudder button is released, the autopilot system will automatically return the rudder to the center position. This makes it easy to realign the vehicle's course without the hassle of manually shifting the lever to the center position.

3. Understand Vertical Movement & Depth

Your vehicle operates in three-dimensional space. However, unlike military submarines, which have separate ballast tanks, this exploratory ROV uses a combination of integrated thrust.

- **Diving Down:** When you push the thrusters above 50% (especially in the 60%–100% range), the hydrodynamic fins and vertical propulsion will work to propel the craft deeper into the ocean abyss. The DEPTH value on the screen

will begin to increase (e.g., from 0.0 m to 100 m, 500 m, or even thousands of meters).

- **Floating Up:**Setting the thrusters to reverse (below 50%) will slowly reduce the depth and pull the vehicle back closer to the surface.
- **Wall Friction Resistance:**If the vehicle gets stuck or stuck to a cliff, the depth gauge may stop moving even if the thrust lever is raised. If this happens, don't push the thrust to 100%! Turn the wheel away from the cliff first, then slowly push.

Driving Safety Protocol:

Maintain Speed on Winding Roads:*The higher the thrust percentage you use, the wider the turning radius the vehicle will need to turn. In narrow, winding troughs, keep the thrust in the range***60% – 70%***so that you have enough reaction time to turn the wheel before the nose of the vehicle hits the coral cliff.*

CHAPTER ⅓
READING SONAR & RADAR TERMINALS

Thousands of meters below the ocean's surface, sunlight is a long-lost concept. The darkness out there is absolute—conventional spotlights can only penetrate a few inches of murky water before their light is absorbed by the dense sea particles.

Your only window to the world is a round, glowing green screen in the center of the cockpit: the Primary Sonar Viewport.

For a seasoned ROV operator, sonar isn't just a graphical diagram. It's the heartbeat of your vehicle. Understanding every contour line, acoustic wave reflection, and telemetry instrument blip is the fine line separating a safe return from becoming a wreck at the bottom of a trench.

1. Understanding the Contour of the Seabed: Safe Paths vs. Coral Walls

The sonar display displays a top-down acoustic topography map of your vehicle's surroundings. You should be able to distinguish terrain conditions within seconds:

- **Open Water Trail (Dark Blue / Black):** This area represents a free water cavity without any solid physical obstructions. It's a safe corridor for the ROV to maneuver, increase propulsion speed, and dive deeper without risk of impact.
- **Cliffs & Solid Coral Formations (Neon Green Gradation):** The circular patterns resembling topographic contour lines in bright to light green indicate dense geological structures—canyon walls, giant sea stalactites, or sheer rock outcrops.
- **Signs of Impact / Friction:** Don't let your vehicle's body overlap with the green area. If the vehicle icon touches the green border, the physics system will slow the vehicle down, and your hull will suffer hull damage.

2. Reading the Vehicle Position on Radar

Right in the center of the radar's concentric circles, you'll see a navigation symbol:

- **Yellow Triangle Icon:**The bright yellow triangle symbol that glows in the center of the screen represents the current physical position of your ROV.
- **Facing Direction (Heading Orientation):**The leading edge of the triangle indicates where the nose of the craft and the front camera are pointing. Pressing the [A] or [D] rudder button will cause the triangle to rotate to follow the craft's bow orientation in real time.
- **Position Coordinates (COORD: X:... Y:...):**In the upper left corner of the radar viewport, the system displays the spacecraft's absolute position within the trench sector. Use these coordinates to chart a return path if you get lost in the maze of narrow crevices.

3. Using Active Sonar Efficiently

Next to the control lever there is an ACTIVE SONAR [ON/OFF] button and a SONAR PING RANGE control lever.

- **How Sonar Beams Work:**A circular radar needle sweep emits active acoustic pulses (pings). Each time the sweep sweeps over a cliff or a mysterious object, the sonar reflections will highlight the obstacle's contours on your screen.
- **Ping Range Lever:**You can adjust the sweep distance of the sound waves (between 100 and 1000 meters). A shorter range provides sharp detail in narrow passages, while the maximum range is useful for searching distant crevasse corridors.
- **The Golden Rule of Silence (Acoustic vs Predatory):**Active Sonar is both a blessing and a curse. Leaving it on is like turning on a loudspeaker in the middle of a jungle filled with hungry predators. Turn the sonar on for 1–2 turns to read the road map ahead, then quickly turn it back off (ACTIVE SONAR [OFF]) as you slip into dangerous depths.

4. Monitoring Vital Telemetry Instrument Box

Around the sonar viewport, there is a vehicle status information panel that you must check periodically:

1. **HULL INTEGRITY (Hull Condition):**
    - Was at 100% at launch.

    - This value will be eroded if you crash into a rock wall or are bitten by a predator. If it reaches 0%, the immense ocean pressure will crush the craft (implosion), destroying your entire mission.
2. **POWER (Battery Backup):**
    - Energy supply for all propulsion systems, lighting, and navigation computers.
    - Running thrusters hard or continuously turning on active modules will drain the battery more quickly. If the battery is completely depleted (0%), the spacecraft will experience permanent signal loss (Signal Loss).
3. **DEPTH & PRESSURE (Hydrostatic Depth & Pressure):**
    - Displays the vehicle's cruising depth in meters (m) and the hydrostatic pressure calculation in MegaPascal (MPa).
    - The deeper you dive, the more pressure the environment increases exponentially. This requires a stronger hull structure and extra vigilance for anomalies at extreme depths.
4. **Descend Rate & Tether Status:**
    - Shows the vehicle's vertical descent rate per second and the stability of the fiber optic cable connecting the surface station. Ensure the cable indicator always shows OK [STABLE].

CHAPTER⅔
FINDING≤ ≤SAMPLING≠ANOMALIES

Amidst the freezing darkness of the abyss, a faint shimmer in the distance is both captivating and dangerous. Hidden beneath it are billions of dollars' worth of scientific treasures: the tissues of ancient organisms capable of withstanding pressures of thousands of atmospheres, high-density mineral crystals never before found in continental crust, and the black debris left by previous expeditions that failed to return to the surface.

Collecting these samples is the primary reason your expedition is funded. But deploying a mechanical arm in the ocean trench is more than just picking up rocks—it's a game of concentration where one second of carelessness could turn your probe from hunter to prey.

1. Understanding the Types of Anomalous Signals

As the probe travels down the trough, the detection instruments will pick up acoustic and visual signatures of anomalies scattered across the abyssal floor:

- **Biomatter / Organic Matter (Green Dot):**
  - Biological tissue samples from deep-sea flora or microorganisms.
  - **Mark:**15 Research Points (RP).
  - **Characteristics:**It's commonly found at shallow depths (0–1,000 meters). The extraction process takes only 2 seconds and produces minimal noise. Ideal for start-up operators looking to securely raise initial capital.
- **Rare Ore / Rare Mineral (Bright Yellow Dot):**
  - High metal content rocks formed by hydrothermal vent activity.
  - **Mark:**50 Research Points (RP).
  - **Characteristics:**Requires minimal maneuvering to approach cliff fractures. Produces standard mechanical vibrations upon extraction.
- **Unknown Anomaly (Mysterious Signal):**
  - Anomalous objects with compositions that are not readable by the initial database.
  - **Mark:**150 Research Points (RP).

    - **Characteristics:** Highly valuable and usually located in deeper zones, it provides a significant boost in research points to unlock advanced booster technology.
- **Black Box / Expedition Data (Double Purple Metal Wreckage):**
    - The black box data recorder from the previous ROV was crushed by pressure or a monster.
    - **Massive Value:** 500 Research Points (RP).
    - **Extreme Danger Level:** Breaking the Black Box encryption requires a full drill for 10 seconds. During this process, the mechanical arm is locked, the spacecraft cannot maneuver, and the sound of the mechanical drill raises the noise level to maximum (NOISE LEVEL MAXIMUM). Retrieving the Black Box without a decoy is premeditated suicide.

2. Procedure for Locking Target & Deploying Sampling Arm

To capture anomalies, follow these operational steps:

1. **Approach the Target Until Entering the Effective Radius:** The ROV's hydraulic arm has a range of 50 units. Using the sonar radar, aim the vehicle's triangular bow toward the anomaly until the object is directly in front of the ROV's nose.
2. **Stop the Ride:** Lower the THRUSTER OUTPUT lever back to neutral (50%) using the [S] key. Extracting while accelerating risks sending the vehicle flying into the cliff.
3. **Press Extract Button:** Click the SAMPLING ARM [EXTRACT] button on the control panel.
4. **Monitor Extraction Cycle:**
    - The button will change status to grey with the words ARM ACTIVE [DEPLOYED].
    - The vehicle's motor system will freeze momentarily while mechanical clamps secure the specimen into the storage tube.
    - Once the cycle is complete, the button will become active again and say SAMPLING ARM [EXTRACT] and the log window will confirm: ANOMALY SECURED.

Extraction System Warning: *If you press the extraction button when there are no anomalies within a 50-unit radius, the system log will display a failure message: EXTRACTION FAILED: NO TARGET IN RANGE. Don't waste time and energy pressing buttons haphazardly!*

3. Securing Research Points (RP)

Each sample successfully clamped by the mechanical arm will be immediately recorded into the vehicle's tactical memory:

- **Auto Tactical Catalog:** Sample data (Specimen ID, Discovery Depth, Anomaly Type, and Hull Condition at time of collection) is directly recorded into a SQLite database.
- **Research Point Vault (RP Vault):** Collected research points are used as the main currency to purchase components in the ACCESS HANGAR.
- **Keys to Success (Data Persistence):** Points you collect in the depths of the ocean are only temporarily stored in the ROV's memory. If the ROV's hull is destroyed (implosion), they will be lost along with the ROV's wreckage. Make sure you have enough battery life and power to disconnect (DATA LINK [DISCONNECT]) to permanently save all RPs.

Operator Survival Rules:

Know When to Take, Know When to Run: *If your sonar detects the roar of an approaching predator,* **cancel the intention to extract anomalies** *, especially the Black Box. Being stuck in the extraction animation for 10 seconds while a predator lunges from the darkness is a sure guarantee of destruction. Protect the spacecraft first, retrieve samples later.*

CHAPTER¼
PREDATOR‹ ‹SURVIVAL⁄PROTOCOL

At depths exceeding a thousand meters, the atmosphere of the trench changes drastically. The once-steady hum of your ROV's engines now feels like a desperate whisper in the void. The water temperature plummets to near freezing, and the darkness beyond is no longer simply the absence of light—it's a living entity.

You soon realize one terrible reality: you are not alone down here.

In the deepest trenches, ancient abyssal predators—often nicknamed Leviathan—roam. They evolved for millions of years in perpetual darkness without eyes. They don't need sight to find prey. In a world without light, sound is their eye, and every vibration of your spacecraft's propeller is an invitation to dinner.

1. Understanding Sound Mechanics & Noisy State

Your survival depends on how well you control the vehicle's noise. The Predator's intelligence system operates purely on acoustic detection and water vibrations.

Your ride will enter a Noisy State and immediately trigger its monster hunting instinct if you do the following:

- **Brutally Pushing the Pusher:** Moving the THRUSTER OUTPUT lever above 65% (more than 15% thrust) causes the propeller blades to cut through the water at high speed, creating water cavitation that produces a sharp, distant buzzing sound.
- **Leaving Active Sonar On:** Active sonar waves bounce high-frequency acoustic pulses in all directions. These waves reflect beautifully on your radar screen, but to a predator's ears, they sound like a blaring siren calling them directly to your location.
- **Running Heavy Extraction Drilling:** The sample extraction process, specifically the Black Box dissection, triggered the roar of the hydraulic drill that forced the spacecraft's noise level to spike to maximum levels for a full 10 seconds.

Silent Mode (Silent Running):

When you realize there is a threat nearby, immediately implement the silent protocol:

1. Lower the thrust to neutral (50%) or keep it in the slow range (52%–58%).
2. Turn off Active Sonar (ACTIVE SONAR [OFF]).
3. Let the vehicle glide slowly following inertia without adding any new thrust.

2. Recognize the signs of predators appearing

Predators won't attack without warning. An alert operator can detect danger before a fatal attack occurs:

- **Audio Warning (Distant Roar):**When diving to depths of 1,000 meters or more, an external hydrophone microphone will occasionally pick up a low, heavy rumbling sound from the abyss. The more frequent and louder the rumbling, the closer the monster's patrol perimeter is to your position.
- **Mobile Radar Contact:**On the sonar screen, the predator will be captured as a contact signal that is actively moving at high speed cutting through the contour lines of the coral, approaching your vehicle's triangle icon.
- **Visual Feed Glitch & Shadows:**If you monitor the front camera, the murky water will churn violently just before the attack. You might catch a glimpse of a giant silhouette passing quickly across the floodlights just before impact.
- **Gastric Shock:**A single bite or strike from a predator can destroy the hull integrity by tens of percent in an instant, accompanied by the deafening sound of an emergency alarm siren in the cockpit.

3. Overcoming Magnetic Signal Interference (Glitch & Distortion)

Besides the physical threat of fangs and jaws, extreme depths harbor magnetic field anomalies that often accompany the movements of ancient seabed entities:

- **Distorted Field Indications:**When the probe gets too close to a mysterious entity or magnetic pocket, the ROV's computer system will experience magnetic interference. Text in the log terminal will begin to become scrambled and corrupted:

  W@RN!NG: C#IT1CAL M@GN3TIC INT3RFERENC3... 5YST3M M4LFUNCT1ON!

- **Visual Impact:**The sonar and front camera viewports will be filled with static lines, random pixel block distortion (glitch blocks), and decreased contour mapping accuracy.
- **Recovery Protocol:**
    - **Do not panic:**Do not immediately pull the steering lever or thruster randomly because this distortion can make you crash into an unseen coral cliff.
    - **Keep the Course Steady:**Maintain constant speed and steer the vehicle away from the epicenter of the disturbance.
    - **Wait for Recalibration:**Once the probe exits the magnetic field radius, the automatic tracking system will restore telemetry with a confirmation message: SYSTEM RESTORED: MAGNETIC INTERFERENCE CLEARED.

Golden Rules of Survival:

Eyes Closed, Ears Open: *In the deep ocean trenches, predators tirelessly search for panicked prey. If you hear a rumbling sound or see any unusual signs on your screen, the best course of action isn't to flee at lightning speed—you'll never be able to outswim the ruler of the deep. Stay still, turn off all active systems, and let the darkness swallow your tracks until the danger has passed.*

CHAPTER¾USE€OF¥WEAPONS° £EMERGENCY≤ DEFENSE

When silent mode fails and a thunderous roar shatters the stillness of the trench directly behind your vehicle, diplomacy with the underwater world is officially over. Faced with the building-sized jaws of a monster hurtling through the darkness at terrifying speed, escape using standard thrusters is futile.

Your only chance of survival lies in your active tactical defense console. The ROV has been equipped with two emergency countermeasure modules: an acoustic decoy and an electromagnetic pulse. These weapons aren't designed to kill, but rather to buy you a few precious seconds to escape death's clutches.

1. Launch Acoustic Decoy (Hotkey [Q])

Acoustic Decoys are your first line of defense when predators start to lock onto the ROV.

- **How Bait Works:**Upon launch, the bait tube ejects from the vehicle's body and begins emitting a very intense pulse of loud sound. To blind deep-sea predators, this bait is far louder and more attractive than your ROV's engines. The predator will immediately shift direction and pounce on the bait.
- **How to Launch:**
    - Press the [Q] key on the keyboard for instant launch, or
    - Click the DEPLOY DECOY [Q] button in the bottom right control panel.
- **Cooldown Time:**After launching, the module takes time to recharge the next launch tube. The button will turn gray and display a countdown timer (e.g., DECOY CD: 15s). During this time, you cannot launch backup decoys.
- **When is the Best Time to Use It?**Launch the decoy as soon as you see a radar contact approaching or when you're forced to drill the Black Box. Once the decoy is launched and the predator is pursuing it, immediately steer the craft in the opposite direction with low engine speed.

2. Detonate an EMP Shockwave (EMP Burst / Hotkey [E])

If the predator is already within striking distance and its bite is seconds away, the decoy won't have time to work. Your last resort is an extreme measure: detonating an EMP (Electromagnetic Pulse).

- **How EMP Works:** The high-power capacitor module will release a massive radial electromagnetic frequency shockwave around the probe. This wave will disrupt the ancient predator's sensitive electroreceptor organs, severely disorienting it, aborting its attack, and forcing it to retreat momentarily.
- **How to Explode:**
  - Press the [E] key on the keyboard, or
  - Click the orange FIRE EMP [E] button on the interface.
- **Long Wait Time:** EMP draws a significant amount of power from the compressor. The button indicator will enter a recharge mode that takes a considerable amount of time (EMP CD: ...s). Don't waste EMP if the threat isn't truly fatal.

3. Consequences of System Blackout Mode

Using an EMP is a double-edged sword. An energy blast capable of crippling a giant predator will also damage your own spacecraft's electronic circuitry from within.

- **Total Blindness (NO SIGNAL):** Moments after the EMP button is pressed, the entire spacecraft subsystem will go into blackout mode. The forward visual camera screen will completely black out, displaying blurry white text reading NO SIGNAL, and the sonar radar will lose its reading of the cliff contours.
- **Temporary Paralysis:** Rudder and thrust controls will become stiff for a few seconds until the backup system successfully reboots the spacecraft's computers.
- **Blackout Strategies:**
  - Before pressing [E], make sure the nose of the vehicle is not facing directly toward the sharp cliff face. The sliding inertia during a blackout could cause the ROV to hit the reef.
  - Remain calm in the control seat. Listen for telemetry signals; once the computer system recovers and the radar screen flashes green again, immediately turn around and move away from the blast area before the predator regains consciousness.

Tactical Usage Guidelines:

Layered Defense Pyramid:*Never immediately discharge an EMP pulse if the threat is still outside the radar's perimeter.***Decoy [Q]***while the monster is still in the distance to divert its route. Save***EMP [E]***purely as a last life-saving button when your vehicle's hull is on the brink of collapse.*

CHAPTER⅛ EVACUATION⇵PROCEDURES≤ ≤STORING≤ RESULTS

At the bottom of the abyss, the line between glorious success and utter ruin is often determined by just one thing: knowing when to quit.

The thrill of a storage tank filled with rare mineral crystals and fragments of classified data is intoxicating. Your inner voice might be telling you to dive a few hundred meters further, searching for that one more anomalous nugget. But creaking mooring cables, dwindling battery indicators, and a giant shadow lurking in the radar's periphery are all clear signs that your time is up. Being a voracious explorer of the ocean floor almost always ends with your name etched on the list of lost vehicles.

A professional operator understands that even the most valuable specimen in the world is worthless if it is smashed to pieces at the bottom of the trough with the remains of your vehicle.

1. Determining the Best Time to Disconnect (Hit and Run Tactics)

The main philosophy of the ABYSS-ROV expedition is not to clear the entire trench in one dive, but to launch repeated rapid attacks (Hit and Run).

Know the right moment to pull yourself out of the depths:

- **Target Achieved (1–2 Valuable Samples):** If you're just starting an expedition or driving a standard-spec vehicle, securing one Rare Ore or two Biomatter units is more than enough to fund essential base upgrades. Don't delay evacuation if your sample bags are full.
- **Gastric Integrity Reaches Danger Limit (< 40%):** A collision with a sharp coral wall or a single predator's strike can quickly shatter a craft's outer structure. When the hull integrity alarm sounds, a small subsequent impact can instantly trigger a high-pressure implosion.

- **Battery Reserve Low (< 20%):** Without electrical power, the propulsion module will be crippled, the searchlights will be turned off, and the defense systems will be incapable of activating. Forcing exploration with minimal battery life is a surefire way to mission failure (Signal Loss).
- **Defense Module Recharging (Cooldown):** If you've just released a decoy or detonated an EMP, and the predator is still roaming near you, immediately find a safe opening and cut off transmission before your protection time runs out.

Transmission Termination Procedure:

1. Reduce the speed of the vehicle and stabilize the bow position away from the steep coral wall.
2. Click the red DATA LINK [DISCONNECT] button on the right-hand instrument panel.
3. The system will immediately lock the telemetry, emergency pull the vehicle to the surface dock, and record the mission's final status as SUCCESS in the base file.

2. Checking Finding History in Tactical Logs

Every time the spacecraft returns from an expedition, all the telemetry data collected during the dive isn't lost. It's all neatly stored in the base's tactical database.

To check the results of your hard work:

- **Opening the Mission Log Window:** Click the Extract button (or the anomaly log menu) in the corner of the interface while on a surface station. The TACTICAL LOGS: EXTRACTED ANOMALIES window will open.
- **Reading the Expedition Data Column:**
    - **Sample ID:** A unique serial number for each specimen successfully brought home.
    - **DepthFound:** The exact depth (in meters) where the specimen was found. This column helps you map which depth layers are richest in a particular resource.

- **AnomalyType:** Categories of scientific findings, ranging from organic materials (ORG_MATTER), rare minerals (RARE_ORE), unknown anomalies (UNKNOWN), to past flight records (BLACK_BOX_DATA).
- **IntegrityAtExtraction:** Records the percentage of the vessel's hull resistance during the retrieval process. This data provides a valuable assessment of the riskiness of your maneuvers at that depth.

Operator Safety Doctrine:

Points in Hand are Better than Gold in a Monster's Lair: *All Research Points (*The Research Points you collect from anomalies will only be permanently transferred to the base vault if the mission ends successfully. If the spacecraft's hull explodes or its batteries run out on the ocean floor, any unsecured samples will be lost forever to the abyss. Get home while you still can!

CHAPTER⅜⅛HANGAR≠ACCESS≤ ⅛RIDE≤ UPGRADES

The smell of hydraulic oil, the hiss of an air compressor, and the distinct aroma of sea vapor greet you as the maintenance dock ladder is lowered. On the base's hanging dock, your ROV is finally free from the thick waters of the trench. Its exterior is dull, scratched by sharp rocks, and drenched by the cold, high-pressure water.

The first dive is just a warm-up. Deeper into the trench, the cliff walls narrow, the undercurrents become wilder, and ancient predators prowl with fangs capable of crushing ordinary steel. To penetrate the true abyssal darkness, the factory specs won't be enough. You'll need to invest every Research Point (RP) you earn to transform this craft into a formidable deep-sea exploration fortress.

1. Open the Hangar / Upgrades Menu

The ride maintenance base is only accessible while the ride is in a surface dock:

- **Access Requirements:**The primary connection button must be in DATA LINK [CONNECT] status. If the mission is still in progress or the connection is still red ([DISCONNECT]), the base security protocol will lock the gate with the warning: ACCESS DENIED: Cannot open Garage during an active mission.
- **Opening the Garage:**Click the red ACCESS HANGAR [UPGRADES] button on the right-hand side panel. This will open the modification workshop interface, displaying the status of your installed modules and your Research Point balance in the base vault.

2. Selecting Improvement Priorities

Each spare part category plays a crucial role in the survival of your vehicle. Allocating research points wisely will determine how far you can go:

- **Hull Integrity Upgrade:**
    - **Function:**Thicken the titanium protective plate and strengthen the main frame structure of the ROV.

    - **Effect:**Increases the maximum limit of resistance to physical damage when hitting a coral cliff and provides additional tolerance when receiving predator bites before the hull is torn.
    - **Priority:**Very High (Mandatory at the Start). Increasing your hull capacity is the best insurance against losing a ride due to a minor navigational error in a narrow passage.
- **Battery Capacity (Battery Level Upgrade):**
    - **Function:**Replacing standard battery cells with high chemical density power storage cells.
    - **Effect:**Extends the duration of active dives, reduces the rate of power drain when revving engines, and ensures searchlights and radar instruments remain on longer.
    - **Priority:**Height. This is crucial if you plan to venture deep into horizontal troughs to search for high-value anomalies without fear of running out of energy on the way back.
- **Thruster Level Upgrade:**
    - **Function:**Modifying the propeller drive motor and hydrostatic propulsion fins.
    - **Effect:**Provides more agile thrust acceleration, sharper steering response, and more efficient vertical cruising range when diving or floating.
    - **Priority:**Intermediate. This upgrade helps you escape more quickly when being chased by predators, but requires greater control precision to prevent the vehicle from spinning out of control.
- **Acoustic Decoy Module (Decoy Upgrade):**
    - **Function:**Shortens the launcher tube's cooldown time and increases the frequency of sound bait attraction.
    - **Effect:**Allows you to launch decoy tubes more frequently when facing emergency situations in predator patrol zones.
    - **Priority:**Medium to High (For Depths > 1,000m).
- **EMP Burst Upgrade Capacitor:**
    - **Function:**Maximizes the blast radius of electromagnetic shock and accelerates the recovery of internal circuits from blackout effects.
    - **Effect:**Immobilizes predators in a wider area while shortening the blind period of the control system after the explosion.

- **Priority:**Final Specialization. This module requires a significant point investment, ideally prepared before hunting for Black Boxes at the bottom of the deepest trenches.

Recommended Upgrade Flow for Beginners:

1. Raise Hull Level to level 2 first to avoid a silly death from cliff wall friction.
2. Increase the Battery Level so that you have enough cruising time at a depth of 500–800 meters.
3. Purchase the Decoy module before you decide to dive through the 1,000-meter mark into Leviathan's lair.
4. Develop the Thruster and EMP modules when you're ready to hunt for rare anomalies and massively valuable data records.

FAQ

1. Why doesn't the ride move even though the propulsion lever has been raised?There are three main possibilities:

- **The lever position is still at the threshold:**50% is the neutral position. 54% provides only minimal boost (about 8%), making movement very slow. Raise the lever to 70%–100% using the [W] key to see the actual movement.
- **Connection status is not active:**Make sure the DATA LINK button is pressed and is red ([DISCONNECT]). If it still shows [CONNECT], the vehicle's transmission system is not running.
- **Hitting the cliff:**If the vehicle's triangular body is attached to or within the solid green area on the radar, the physical system will restrain the vehicle's speed. Turn the rudder with [A] or [D] away from the reef, then use reverse thrust with [S].

2. Why does my ride always get destroyed quickly when attacked by monsters in the depths?The ancient monster (Leviathan) had no eyes and hunted purely using sound vibrations:

- **Reduce engine speed:**Pushing the thruster above 65% triggers the Noisy State.
- **Turn off Active Sonar:**Leaving the sonar on continuously emits sound pulses that guide the monster directly to your location. Turn it on for 1–2 turns to read the route, then immediately turn it off.
- **Avoid Black Box extraction at the beginning:**Retrieving the black box locks the mechanical arm for 10 seconds and triggers a maximum drill roar that summons predators remotely.

3. How do I save Research Points (RP) so they don't disappear when I die?Research points earned from anomalies are only saved permanently if you manage to return to the surface safely:

- Apply Hit and Run tactics: After securing 1–2 samples, immediately press the red DATA LINK [DISCONNECT] button.
- The mission will conclude with a successful status, and all points will be immediately credited to the base's vault. If the hull explodes or the batteries run out on the seabed, all samples on board the spacecraft will be destroyed.

4. When is the right time to use Acoustic Decoy instead of EMP Burst?Use defenses in stages according to the threat distance:

- **Acoustic Decoy [Q]:**Use it when the monster is still within radar range or swimming closer. This decoy emits a loud noise that diverts the monster's hunting route without damaging your own system.
- **EMP Burst [E]:**Save this as a last resort when the monster is right in front of the ride's nose and ready to bite. An EMP blast instantly incapacitates the monster but blacks out your screen and ride controls for a few seconds.

5. Why can't the ACCESS HANGAR menu be opened?The garage workshop gate is automatically locked for work safety protocols:

- The garage and upgrade menus can only be opened when the ride is on the surface dock in standby status.
- Make sure the main switch is in DATA LINK [CONNECT] status. If the mission is still active, disconnect it before attempting to install new parts.

6. Why is the front camera display so dark and full of floating spots?The visual camera simulates abyssal darkness without direct sunlight. The probe's spotlight only penetrates a short distance in front of the bow, while floating white specks are organic marine particles. The camera's view will begin to reveal the silhouette of coral reefs as the probe approaches a cliff, or the shadow of a predator just before an attack.

7. What does the sudden messed up log text and shaking screen mean?This condition indicates the spacecraft's computer system is experiencing strong magnetic field disturbances. This phenomenon is usually triggered by the spacecraft's proximity to certain geological anomalies or the movement of ancient entities. This disturbance is temporary. Maintain steering direction away from the center of the disturbance until the navigation system automatically recalibrates.

8. Why does the Black Box extraction process take much longer than other samples?The black box holds highly valuable expedition data, amounting to 500 research points. The system requires a full 10 seconds to crack the data vault. During the drilling process, the vehicle cannot be moved, and the sound of the drilling machine triggers the highest noise level. Ensure the surrounding area is clear or prepare decoys before touching this object.

9. Why does the steering wheel always return to the center when the keyboard button is released?The steering system features an automatic balance feature to prevent the vehicle from spinning out of control in narrow trough passages. When the [A] or [D] button is released, the rudder immediately returns to the center position, allowing you to focus on controlling the thruster's momentum without worrying about the vehicle veering off the cliff.

10. What are the consequences if the battery runs out completely while diving?If the power indicator reaches 0%, all operational subsystems, including thrusters, sonar, cameras, and defense modules, will shut down completely. The spacecraft will become paralyzed at depth, triggering a mission failure status due to signal loss. Any specimens not recovered to the surface will be permanently lost along with the spacecraft.

11. Why does the SAMPLING ARM [EXTRACT] button display the message 'EXTRACTION FAILED: NO TARGET IN RANGE'?The mechanical extraction arm has a reach radius limit of 50 units. This failure message appears if the ROV is too far from the anomaly, the object is outside the forward bow reach angle, or the specimen has already been successfully retrieved. Approach the target object until it is almost touching the probe's nose before pressing the extraction button.

12. Why does the Hull Integrity keep decreasing even though there are no monsters around?Physical damage to the hull often doesn't come from attacks, but rather from constant impact or friction with the coral cliff walls. On the radar display, ensure the craft's triangular hull doesn't overlap with the solid green area. Rapid maneuvering in narrow passages without considering the inertia of the water slide will cause the hull to continually be impacted and gradually eroded.

13. What is the difference in value between Biomatter, Rare Ore, and Unknown Anomaly samples?Each specimen category rewards different research points according to its rarity:

- **Biomatter (Green Dot):**Gives 15 RP, most easily found in shallow areas with fast extraction times.
- **Rare Ore (Yellow Dot):**Gives 50 RP, located in the coral crack.
- **Unknown Anomaly:**Gives 150 RP, is at a more challenging depth.
- **Black Box (Double Purple):**Worth a massive 500 RP, but triggers the risk of noise hazard and a 10-second paralysis penalty.

14. How to use the SONAR PING RANGE lever effectively?This lever sets the radar wave sweep radius between 100 and 1,000 meters:

- Use short distances (100–350 meters) when maneuvering in narrow cliff gaps so that the visual contours of obstacles around the vehicle are close, clear, and accurate.
- Use long range (700–1,000 meters) occasionally while in open space to map the continuation of the trough corridor from a distance, then return to medium range to avoid biasing short-range navigation readings.

15. What happens when the vehicle passes the 1,000 meter depth limit?Passing the 1,000-meter mark marks the spacecraft's entry into the deep abyssal zone. At this depth, the hydrostatic pressure increases dramatically, and the prehistoric predator's intelligence system (Leviathan) begins active patrols. Audio hydrophones will begin to periodically pick up low roars, signaling that you must implement silence protocols and disable active radar systems to avoid detecting the spacecraft's presence.

16. What does the TETHER STATUS indicator on the radar screen mean?This indicator monitors the physical integrity of the fiber optic cable (tether) connecting the ROV to the surface control station. A status of OK [STABLE] indicates that the telemetry data transmission, steering control, and visual feed are operating optimally and without interruption. If the vehicle becomes entangled in a complex cliff face or is ejected by an explosion, the cable's stability could be compromised, alerting the operator to immediately free the vehicle to avoid losing transmission.

17. Why does the DECOY or EMP button sometimes not respond when pressed?This condition is generally caused by two operational factors:

- **The system is in recovery period (Cooldown):**Once activated, the defense module requires a recharge period. The button will turn gray and display a countdown timer (e.g., DECOY CD: ...s or EMP CD: ...s). The button will become usable again once it reverts to DEPLOY DECOY [Q] or FIRE EMP [E].
  PAS+ 3
- **Module specifications on the base:**The charging duration and effectiveness of these two emergency weapons depend on the level of the module installed via the Hangar menu.

18. Are the upgrades to the rides in the Hangar permanent? Yes. All component upgrades you purchase in the Hangar—such as hull reinforcements, battery capacity, thrusters, and defense modules—are permanently stored in the base's SQLite database. The level of any parts you've unlocked will not be lost or reset even if the craft is destroyed (imploded) during the next mission.

19. How do I check the history of all specimens that have been secured? When the probe is docked at the base with the connection down, click the Extract button in the bottom right corner. The interface will open a data archive window (Tactical Logs) containing a detailed history table of all anomalies from previous expeditions, including sample number, depth of discovery, anomaly category, and the condition of the probe's hull at the time of collection.

20. Is the mission automatically completed if the vehicle successfully ascends back to a depth of 0 meters? Bringing the float back to 0 meters (by lowering the thrusters below 50%) does not automatically end the expedition simulation. To officially record the mission as a success, stop power consumption, and permanently transfer all accumulated research points to the base vault, you must still press the DATA LINK [DISCONNECT] button.

TIPS

1. Navigation Discipline & Silent Running Mode

- **Avoid Pushing the Maximum Thrust:** Limit the thrust lever to 55%–65% when maneuvering. Pressing the thrust lever above 65% triggers water cavitation (Noisy State), which attracts predators from a distance.
- **Apply the Inertial Coasting Technique:** Seawater has mass and inertial resistance. Give it a short push to accelerate, then return the lever to the neutral (50%) position. Allow the ROV to glide silently without propeller rotation as it passes through tight spaces.

- **Turn Off Active Radar Periodically:** Do not leave the Active Sonar switch on ([ON]). Activate it for 1–2 strokes to read the cliff route ahead, then immediately turn it off ([OFF]) to break the acoustic trail.

2. Smart Sampling & Extraction Tactics

- **Secure Shallow Targets Early in the Game:** Focus your initial dives on Biomatter (15 RP) and Rare Ore (50 RP) samples. These samples are in relatively safe zones with very short extraction times.
- **Special Preparation Before Extracting Black Box:** Retrieving the Black Box (500 RP) requires a full 10 seconds of drilling time, which locks the vehicle and triggers maximum noise. Before pressing the extraction button:

    1. Make sure the bow faces the open escape route, not the cliff wall.
    2. Set up the shortcut key [Q] to launch a decoy.

    3. Release the bait just before starting the drill if a monster roar is detected in the distance.
- **Stop the Ride Before Opening the Arms:** Always slow the ride down to a stop before pressing the extraction button. Extracting while traveling at high speed often causes the ride to bounce off the cliff face and damage the hull's integrity.

3. Layered Defense Scheme

- **Keep Your Distance with Acoustic Decoy [Q]:** Use decoy bait as soon as possible when detecting moving contact on radar. As soon as the bait is released, immediately turn the vehicle at a 90-degree angle in stealth mode to lure predators to chase the decoy signal.

- **Save EMP Burst [E] for Distressed Situations:** Never use an EMP pulse if the enemy is still far away. The EMP triggers a system blackout (Blackout Mode), which disables your screen and vehicle controls for a few seconds. Use the EMP only when the predator's jaws are already in front of the camera.

- **Stabilize Position Before Detonating EMP:** Before pressing the [E] button, ensure the vehicle's bow is not facing directly towards the sheer rock formation. The inertia of the vehicle with the screen completely blacked out (NO SIGNAL) can trigger a fatal collision if the bow faces the cliff.

4. Telemetry Management & Golden Rule: Hit and Run

- **Apply the 40/20 Tolerance Limit:**
    - Immediately disconnect if the hull integrity reaches < 40%.

    - Start moving to the safe zone if the battery reserve (Power) reaches < 20%.

- **Secure Research Points Gradually:** Collecting 200–300 RP and then immediately pressing the DATA LINK [DISCONNECT] button is much more effective than trying to earn thousands of points only to end up lost. Each time the connection is successfully disconnected, all points are added to the base vault without any risk of loss.

5. Hangar Improvement Roadmap Priorities

When accessing the upgrade menu in the surface dock (ACCESS HANGAR), spend research points in the following order of priority:

1. **Hull Reinforcement (Levels 1–2):**Thickening the hull is a top priority so that the vehicle has tolerance to impacts from coral cliffs in narrow passages.
2. **Battery Capacity:**Increases power reserves to extend dive duration at depths of 500–1,000 meters.
3. **Decoy Cooldown:**Speeds up the reloading of the bait launcher before exploring zones above 1,000 meters.
4. **Thruster Efficiency & EMP Module:**Maximize your propulsion power and pulse module when you are ready to hunt for Black Boxes in the deepest trenches.